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Ch.6 Bioenergetics

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

Flowchart comparison task: photosynthesis and respiration

Photosynthesis uses CO_2 and H_2O, absorbs light energy, and produces glucose and O_2. Its light-dependent reactions make ATP and NADPH; the Calvin cycle uses them to reduce CO_2 to sugar.

Cellular respiration uses glucose and O_2, releases CO_2 and H_2O, and transfers energy to ATP. Glycolysis, the Krebs cycle and the electron transport chain oxidize glucose step by step.

Thus the products of photosynthesis are the reactants of respiration, and the products of respiration are the reactants of photosynthesis.

SECTION 1: MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. What is the main process of the dark reaction (Calvin cycle)?

Answer: (c) Adding hydrogen to CO_2 — The Calvin cycle reduces CO_2 using NADPH and ATP to form carbohydrate.

2. Which statement about glycolysis is true?

Answer: (d) It reduces 2 molecules of NAD^+ for every glucose processed — Glycolysis produces 2 NADH, 2 ATP (net) and 2 pyruvate per glucose in the cytoplasm.

3. Which products of thylakoid reactions are consumed in the stroma?

Answer: (d) ATP and NADPH — Light reactions make ATP/NADPH; the Calvin cycle uses them to fix CO_2.

4. Yeast fermentation produces CO_2, ATP and what else?

Answer: (b) Ethyl alcohol — Alcoholic fermentation converts pyruvate to ethanol and CO_2.

5. Conversion of glucose-6-phosphate to fructose-6-phosphate is what?

Answer: (a) Isomerization — An isomerase rearranges the molecule without adding/removing atoms.

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6. In which conversion is ATP produced?

Answer: (b) Succinyl CoA → succinic acid — This Krebs-cycle step produces GTP/ATP by substrate-level phosphorylation.

7. In the textbook ETC yield, $FADH_2$ produces how many ATP?

Answer: (b) Two — The textbook uses the classical yield of 2 ATP per $FADH_2$ (modern estimates are about 1.5).

8. Which molecule is the CO_2 acceptor in photosynthesis?

Answer: (b) Ribulose bisphosphate — RuBP combines with CO_2 in the Calvin cycle, catalysed by Rubisco.

9. Which molecule accepts electrons lost by Photosystem I?

Answer: (a) Ferredoxin — Excited PSI electrons are transferred to ferredoxin and then used to reduce $NADP^+$.

10. Photosystem II replaces lost electrons from what?

Answer: (d) Photolysis of water — Water splitting supplies electrons to PSII and releases O_2 and H^+.

SECTION 2: SHORT QUESTIONS

1. Differentiate between action spectrum and absorption spectrum.

Absorption spectrum	Action spectrum
Shows: Wavelengths of light absorbed by a pigment.	**Shows:** Rate of photosynthesis at different wavelengths.
Measured by: Spectrophotometer/pigment extract.	**Measured by:** O_2 production or CO_2 uptake.
Meaning: Physical light absorption.	**Meaning:** Biological effectiveness of light.

2. How is photosynthesis a redox reaction?

Water is oxidized to O_2 (loses electrons/hydrogen), while CO_2 is reduced to carbohydrate (gains electrons/hydrogen). Therefore photosynthesis involves coupled oxidation and reduction.

3. Which molecule contributes the oxygen in glucose: water or carbon dioxide?

The oxygen atoms in glucose come from **CO_2**. The O_2 released during photosynthesis comes from water.

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4. State the role of CO_2 in photosynthesis.

CO_2 is the carbon source. It is fixed by RuBP in the Calvin cycle and reduced to form sugars.

5. Define electron transport chain.

An electron transport chain is a series of membrane carriers that pass electrons from donors to acceptors. Energy released is used to pump protons and make ATP by chemiosmosis.

6. What is glycolysis?

Glycolysis is the cytoplasmic breakdown of one glucose into two pyruvate molecules, producing a net 2 ATP and 2 NADH. It does not require oxygen.

7. What is the main structural difference between chlorophyll a and b?

Chlorophyll a has a methyl (–CH_3) group at a specific porphyrin side position, while chlorophyll b has an aldehyde (–CHO) group there. This shifts their absorption spectra.

8. How can a cell make ATP by substrate-level phosphorylation?

A phosphate group is transferred directly from a phosphorylated substrate to ADP, forming ATP. Examples occur in glycolysis and the Krebs cycle (succinyl CoA → succinate).

9. Can pyruvic acid enter the Krebs cycle as such? If not, what change occurs?

No. Pyruvate is first converted to acetyl CoA by oxidative decarboxylation: CO_2 is removed, NAD^+ is reduced to NADH, and coenzyme A attaches. Acetyl CoA then enters the Krebs cycle.

10. Differentiate between C3 and C4 photosynthesis.

C3	C4
First stable product: 3-phosphoglycerate (3C).	**First stable product:** Oxaloacetate/malate (4C).
CO_2 acceptor: RuBP.	**CO_2 acceptor:** PEP in mesophyll; CO_2 released to RuBP in bundle sheath.
Anatomy: No Kranz anatomy.	**Anatomy:** Kranz anatomy common.
Photorespiration: Higher in hot/dry conditions.	**Photorespiration:** Reduced by CO_2 concentration mechanism.

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SECTION 3: LONG QUESTIONS

1. What are photosynthetic pigments and what is their role in light absorption/conversion?

Photosynthetic pigments are molecules in thylakoid membranes that absorb light energy. The main types are chlorophylls (a, b, c, d) and carotenoids (carotenes and xanthophylls); phycobilins occur in red algae and cyanobacteria.

- **Chlorophyll a:** main reaction-centre pigment; converts absorbed light into chemical energy by exciting electrons.
- **Accessory pigments:** chlorophyll b, carotenoids and phycobilins absorb additional wavelengths and transfer energy to chlorophyll a.
- **Photoprotection:** carotenoids dissipate excess energy and protect the photosynthetic apparatus from damage.

Energy transfer order in plants: carotenoids → chlorophyll b → chlorophyll a. The excited electrons then enter the electron transport chain.

2. How are the absorption spectra of chlorophyll a and b different?

Both chlorophylls absorb mainly violet-blue and orange-red light and reflect green, but their peaks differ because of side-group chemistry.

- Chlorophyll a has a methyl ($–CH_3$) group on the second pyrrole ring and absorbs best around 430 nm and 662 nm.
- Chlorophyll b has an aldehyde (–CHO) group at the same position and absorbs best around 453 nm and 642 nm.

These complementary peaks broaden the range of light usable for photosynthesis. The action spectrum of photosynthesis closely matches the combined absorption spectrum of the pigments.

3. Describe how pigments are organized in the thylakoid membrane.

Pigments are organized into clusters called photosystems embedded in the thylakoid membrane. Each photosystem has a light-harvesting antenna complex and a reaction centre.

- **Antenna complex:** many pigment molecules capture photons and pass excitation energy to the reaction centre.
- **Reaction centre:** contains chlorophyll a molecules that pass high-energy electrons to a primary electron acceptor.
- **Photosystem II:** reaction-centre chlorophyll P680 absorbs best at 680 nm.
- **Photosystem I:** reaction-centre chlorophyll P700 absorbs best at 700 nm.

Electron carriers between the photosystems form the electron transport chain, linking light capture to ATP and NADPH formation.

4. How can the role of water in photosynthesis be shown experimentally?

Use the heavy oxygen isotope ^{18}O as a tracer.

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- Group 1: supply $H_2{}^{18}O$ + normal CO_2 → released O_2 is $^{18}O_2$ (labelled).
- Group 2: supply normal H_2O + $C^{18}O_2$ → released O_2 is normal $^{16}O_2$ (unlabelled).

This proves that the oxygen evolved during photosynthesis comes from water, not carbon dioxide. Water is split (photolysis) to supply electrons and H^+, releasing O_2 as a by-product.

5. What events capture light and convert it to chemical energy in light-dependent reactions?

Light absorbed by PSII (P680) → Water photolysis supplies electrons, H^+ and O_2 → Electrons pass ETC (PQ → cytochrome → PC), pumping H^+ → ATP formed by chemiosmosis → Light absorbed by PSI (P700) re-excites electrons → Ferredoxin → $NADP^+$ reductase → NADPH

This non-cyclic electron flow (Z-scheme) produces ATP, NADPH and O_2. The ATP and NADPH are used in the Calvin cycle to reduce CO_2 to sugar.

6. Illustrate cyclic photophosphorylation.

PSI absorbs light → Electrons excited to primary acceptor/ferredoxin → Electrons cycle back through ETC to P700 → Proton gradient forms → ATP synthesized by chemiosmosis

Only PSI is involved. Electrons return to the reaction centre, so no NADPH is produced and no water is split, hence no O_2 is released. Cyclic flow generates extra ATP when the Calvin cycle slows and NADPH accumulates.

7. Describe light-independent reactions and illustrate the Calvin cycle.

The Calvin cycle occurs in the stroma and uses ATP and NADPH from light reactions to fix CO_2 into sugar. It has three phases:

- **Carbon fixation:** Rubisco combines 3 CO_2 with 3 RuBP (5C) to form 6 molecules of 3-PGA (3C).
- **Reduction:** 6 ATP and 6 NADPH convert 6 3-PGA into 6 G3P; one G3P leaves to form glucose/other carbohydrates.
- **Regeneration:** 5 G3P are rearranged using 3 ATP to regenerate 3 RuBP, so the cycle continues.

Overall requirement for one glucose

$6CO_2 + 12NADPH + 18ATP \rightarrow 1$ glucose (net) $+ 12NADP^+ + 18ADP + 18Pi$

8. What happens to glucose in anaerobic respiration and how do organisms modify end products?

Glucose is partially broken down by glycolysis to two pyruvate molecules, yielding a net 2 ATP and 2 NADH. Without oxygen, pyruvate is reduced to regenerate NAD^+ so glycolysis can continue.

- **Alcoholic fermentation:** in yeast and some bacteria, pyruvate → ethanol + CO_2.
- **Lactic acid fermentation:** in animal muscles and some bacteria, pyruvate → lactic acid.

Only about 2% of glucose energy is captured; most remains in ethanol or lactate.

9. How is glucose broken down to pyruvic acid in glycolysis?

Preparatory phase (uses 2 ATP)

Glucose → glucose-6-phosphate → Glucose-6-phosphate → fructose-6-phosphate → Fructose-6-phosphate → fructose-1,6-bisphosphate → Split into two G3P molecules

Payoff phase (produces 4 ATP + 2 NADH)

G3P oxidized → 1,3-bisphosphoglycerate (NADH formed) → Substrate-level phosphorylation → 3-PGA + ATP → 3-PGA → 2-PGA → PEP → PEP → pyruvate + ATP

Net per glucose: 2 pyruvate + 2 ATP + 2 NADH. Glycolysis occurs in the cytosol and does not require oxygen.

10. Describe how the Krebs cycle completes oxidation of glycolytic products.

Pyruvate is first converted to acetyl CoA (releasing CO_2 and forming NADH). Acetyl CoA (2C) combines with oxaloacetate (4C) to form citrate (6C).

Citrate → α-ketoglutarate (CO_2 + NADH) → α-ketoglutarate → succinyl CoA (CO_2 + NADH) → Succinyl CoA → succinate (ATP/GTP) → Succinate → fumarate ($FADH_2$) → Fumarate → malate → oxaloacetate (NADH)

Per acetyl CoA: 2 CO_2, 3 NADH, 1 $FADH_2$ and 1 ATP are produced. Oxaloacetate is regenerated, so the cycle continues. The reduced coenzymes then feed the electron transport chain, completing oxidation of glucose to CO_2.

11. Explain the passage of electrons through the electron transport chain.

NADH and $FADH_2$ donate electrons to carriers in the inner mitochondrial membrane.

$NADH/FADH_2$ → coenzyme Q → CoQ → cytochrome b → cytochrome c → Cytochrome c → cytochrome a/a_3 complex → Electrons passed to O_2, forming H_2O

As electrons move down the chain, energy is released and used to pump H^+ from the matrix into the intermembrane space. The resulting proton gradient drives ATP synthase. The textbook uses classical yields: 3 ATP per NADH and 2 ATP per $FADH_2$, giving a net total of about 36 ATP per glucose.

12. Define chemiosmosis and relate it to the ETC.

Chemiosmosis is ATP synthesis driven by the diffusion of H^+ down an electrochemical gradient through ATP synthase.

The ETC uses energy from electrons to pump H^+ across a membrane: from matrix to intermembrane space in mitochondria, and from stroma to thylakoid lumen in chloroplasts. The stored gradient has potential energy. As H^+ flows back through ATP synthase, the enzyme phosphorylates ADP to ATP.

Thus the ETC creates the gradient, and chemiosmosis couples that gradient to ATP production.

13. How do proteins and fats enter cellular respiration?

- **Proteins:** hydrolysed to amino acids; after deamination (removal of amino group), carbon skeletons are converted to pyruvate, acetyl CoA or Krebs-cycle intermediates.
- **Fats:** hydrolysed to glycerol and fatty acids. Glycerol is converted to glyceraldehyde-3-phosphate and enters glycolysis. Fatty acids undergo β-oxidation to acetyl CoA, which enters the Krebs cycle.

Fats yield more energy per gram because they contain many C–H bonds and are highly reduced.

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14. Define photorespiration and explain why it shows photosynthesis is not perfect.

Photorespiration is a light-dependent process in which Rubisco binds O_2 instead of CO_2, consuming energy and releasing CO_2 without producing sugar.

> RuBP + O_2 → phosphoglycerate + phosphoglycolate → Phosphoglycolate → glycolate (chloroplast → peroxisome) → Glycolate → glycine → serine (mitochondrion, CO_2 released) → Serine → glycerate → phosphoglycerate (back to Calvin cycle)

It is favoured by high temperature, high O_2 and low CO_2 (closed stomata). C3 plants may lose 25–50% of fixed carbon, reducing yield. This wasteful side reaction shows that the carbon-fixing system is not perfectly efficient.

15. What are the effects of temperature on the oxidative activity of Rubisco?

At higher temperatures, Rubisco's oxygenase activity increases relative to its carboxylase activity. Also, hot/dry conditions cause stomata to close, lowering internal CO_2 and raising O_2.

Both effects favour photorespiration over carbon fixation, so net photosynthesis falls. In tropical climates above about 28°C this can severely reduce agricultural yields. C4 and CAM plants have adaptations to reduce this problem.

16. How is C4 photosynthesis an adaptation to reduce photorespiration?

C4 plants (e.g., maize, sugarcane, sorghum) initially fix CO_2 in mesophyll cells using PEP carboxylase, forming a 4-carbon acid (oxaloacetate → malate).

> CO_2 + PEP → oxaloacetate (mesophyll) → Malate transported to bundle-sheath cell → Malate → pyruvate + CO_2 → CO_2 concentrated near Rubisco → Calvin cycle → Pyruvate returns to mesophyll and regenerates PEP (uses ATP)

Bundle-sheath cells maintain high CO_2 concentration, suppressing Rubisco's oxygenase activity. Thus C4 photosynthesis minimizes photorespiration in hot, bright environments, at the cost of extra ATP.

INQUISITIVE QUESTIONS

1. Why does cellular respiration release energy more efficiently than fermentation?

Respiration completely oxidizes glucose to CO_2 and H_2O and uses the ETC to capture much energy as ATP (about 30–38 ATP in textbook accounts). Fermentation only partially breaks down glucose and yields just 2 ATP, leaving much energy in ethanol or lactate.

2. Why is converting glucose to ATP more efficient than burning glucose directly?

Cellular respiration releases energy in small enzyme-controlled steps and stores it in ATP. Burning releases energy suddenly as heat/light, which cells cannot use efficiently and would damage tissues.

3. Why would disruption of photosynthesis or respiration affect global carbon and oxygen cycles?

Photosynthesis removes CO_2 and releases O_2, while respiration consumes O_2 and releases CO_2. These processes balance atmospheric gases and move carbon through ecosystems. Disrupting either changes CO_2/O_2 levels, climate and energy flow in food webs.

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